MAFS5440 Final Project: M5 Forecasting

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MAFS5440

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Project Overview

- The M5 competition aims to forecast daily sales for the next 28 days(till 22nd May 2016), and to make uncertainty estimates for these forecasts, utilizing historical daily sales data from Walmart stores.**We joined both Accuracy and Uncertainty competition.**
- Accuracy estimates the unit sales of Walmart retail goods for the next 28 days.
- Uncertainty estimates the uncertainty distribution of Walmart unit sales for the next 28 days.

Data

- the dataset involves the unit sales of 3,049 products, classified in **3 product categories** and **7 product departments**, in which the above-mentioned categories are disaggregated. The products are sold across **10 stores**, located in **3 States**.
- The historical data range from 2011-01-29 to 2016-06-19, which means the products have a selling history of 1,941 days(test data of h=28 days not included).
- The train-val-test split is shown as follows:
  - Training set: 2011-01-29 to 2016-04-24 (d_1 to d_1913)
  - Validation set: 2011-04-25 to 2016-05-22 (d_1913 to d_1941)
  - Testing set: 2011-05-22 to 2016-06-19 (d_1941 to d_1969)

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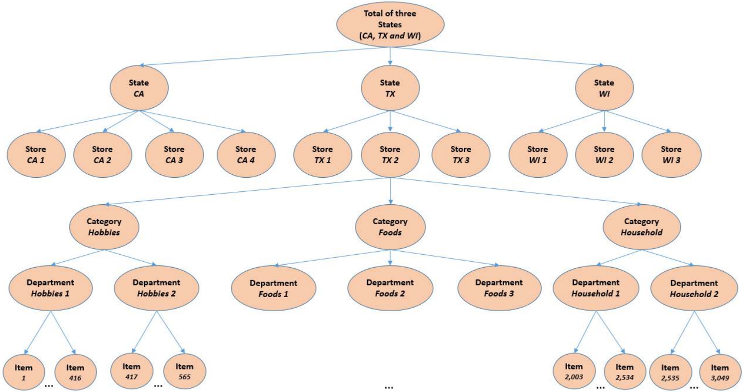
Data overview



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Features

In conclusion, there are 5 categories of features applied in our project:

- sales features
- price features
- calendar features
- lag features
- mean encoding features

Features overview

Below is the overview of features we applied in our project:

Type	Column name
sales features	sales
price features	sell price price aggregations(max,min,mean,std) price_norm price nunique price momentums(week,month,year)
calendar features	event related features day,week,month,month,year day of week, week of month whether weekend
lag features	sales lag features(1-14) rolling mean, std features(7,14,30,60,180)
mean encoding features	enc_store_id_dept_id_mean enc_store_id_dept_id_std enc_item_id_state_id_mean enc_item_id_state_id_std

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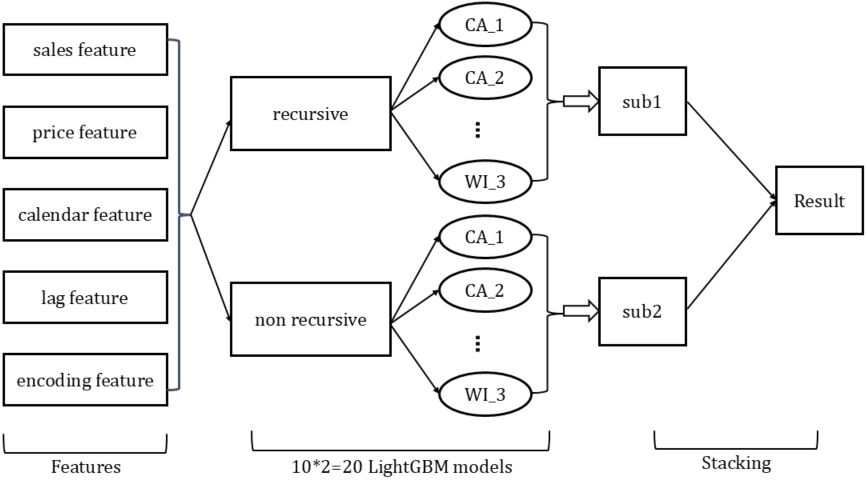
Modeling Strategy

- We apply recursive and non-recursive strategy separately, and we ensemble the two part as our final model strategy
- For each strategy, we divide the training data into groups by store_id, and apply LightGBM for each group to train and predict.
- As there are 10 stores in dataset, we have 10 models for each strategy. Therefore, we have 10*2=20 LightGBM models for our forecasting approach.

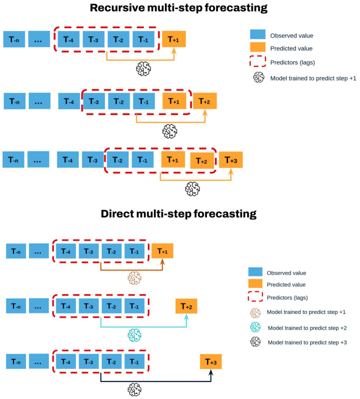
Flowchart


sales feature
price feature
calendar feature
lag feature
encoding feature
recursive
non recursive
CA_1
CA_2
WI_3
CA_1
CA_2
WI_3
sub1
sub2
Result
Features
10*2=20 LightGBM models
Stacking

Recursive Model v.s. Non-Recursive Model

- For recursivce model, The model is trained in a loop for each store, and predictions are made recursively for each day (i.e., the model uses the predictions from previous days to inform future predictions).
- For non-recursive model, The model is trained on historical data for each store and validated without using previous predictions. The predictions are made for a fixed validation window without any feedback from earlier predictions.

Recursive Model v.s. Non-Recursive Model(Con't)



LightGBM

We choose LightGBM with Tweedie objective function as our only model. The parameters we choose are as follows:

```
############################ Model params
##################################################################################
import lightgbm as lgb
lgb_params = {
                    'boosting_type': 'gbdt',
                    'objective': 'tweedie',
                    'tweedie_variance_power': 1.1,
                    'metric': 'rmse',
                    'subsample': 0.5,
                    'subsample_freq': 1,
                    'learning_rate': 0.015,
                    'num_leaves': 2**11-1,
                    'min_data_in_leaf': 2**12-1,
                    'feature_fraction': 0.5,
                    'max_bin': 100,
                    'n_estimators': 3000,
                    'boost_from_average': False,
                    'verbose': -1,
                }
```

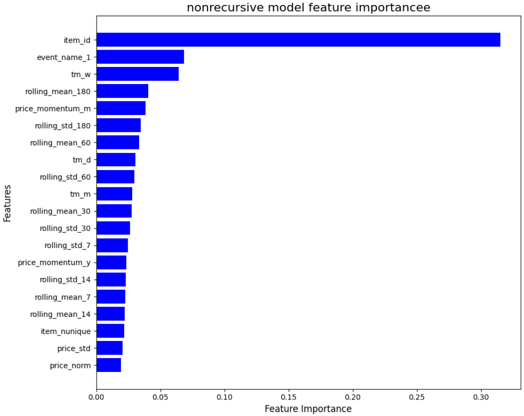
Ensembled Result

After we ensembled recursive and non-recursive models, the Kaggle result is as follows:

Submission and Description	Private Score	Public Score	Selected
submission_final.csv Complete (after deadline) · 1d ago · ensembled model	**0.55095**	**0.38660**	☐

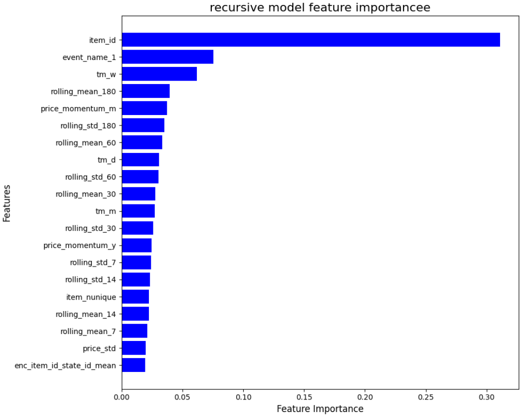
Feature Importance

The feature importance for recursive LightGBM models are:



Feature Importance(Con't)

The feature importance for non-recursive LightGBM models are:



Feature Importance(Con't)

Analyzing the feature importance graphs, there are two catogories of features that have high importance in our model:

- lags features: rolling mean, rolling std.
- calendar features: week, day, month, event.

Although price features and mean encoding features also contributes to the model, we can find that major features are mainly lags features and calendar features.

Review of Data Structure

After examining the sample submission data, we realized that we need to make quantile predictions for each level. The level 12 is the bottom-level, which is the only non-aggregated one. The level 1 is the top-level, which is the only all-aggregated one.

Level id	Aggregation Level	Number of series
1	Unit sales of all products, aggregated for all stores/states	1
2	Unit sales of all products, aggregated for each State	3
3	Unit sales of all products, aggregated for each store	10
4	Unit sales of all products, aggregated for each category	3
5	Unit sales of all products, aggregated for each department	7
6	Unit sales of all products, aggregated for each State and category	9
7	Unit sales of all products, aggregated for each State and department	21
8	Unit sales of all products, aggregated for each store and category	30
9	Unit sales of all products, aggregated for each store and department	70
10	Unit sales of product *x*, aggregated for all stores/states	3,049
11	Unit sales of product *x*, aggregated for each State	9,147
12	Unit sales of product *x*, aggregated for each store	30,490
Total		**42,840**

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Workflow

- Having the median from the Accuracy track as a starting point for level 12, with simple aggregations we obtain the median for the other 11 levels.
- We need to extract quantile estimation from normal distribution. We first assign a coefficient to different levels, then we map original quantiles to a set of new quantiles by the coefficient. We could calculate CDF of new quantiles in standard normal distribution.
- After normalizing we get the ratios for each level. The prediction interval is calculated by multiplying ratios and accuracy results.
- we know that normal distribution is a too strict assumption and we should give some space to our estimation. We finally widen our quantile estimates by 0.5%.

Result

The Kaggle result for Uncertainty competition is as follows:

Submission and Description	Private Score	Public Score	Selected
submission_uncertainty_final.csv Complete (after deadline) · 15s ago · Uncertainty final submission	**0.22243**	**0.14297**	☐

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Further Improvements

- **Accuracy**:
  - Dividing training data in different ways.
  - Introduce weekly models.
  - Try other models.
- **Uncertainty**:
  - Customized distribution.
  - Tune the parameter.

Conclusion

We have learned a lot during the process of completing the final project. There are two key takeaways we learn from the final project

- **Apply simple but practical model:** Although complicated would generate better results most of the time. We need to find the most simple but effective model for our problem instead of using fancy models without careful thoughts.
- **Learn from the Kaggle community:** During the process, we have gone through many discussions and experience from medal winners in Kaggle. Many of the ideas of feature engineering and model strategy are learned and combined from those excellent and kind Kaggle players. We want to express our sincere thanks to such a insghtful community.